A Skill for Large Language Models to Evaluate Pathology Report Quality

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Introduction

Complete pathology report is essential for patient management and accreditation purposes. Although synoptic and structured reports are becoming more standard, using free-text reports is still common. We have developed a Skill for Large Language Models to extract information and validate report quality.

Material and methods

An AI quality assurance tool was built using Claude's skill-builder framework. It was designed to evaluate ICCR required elements, TNM validation logic, macroscopic features, interreport integrity.

One hundred anonymized Turkish-language colorectal resection reports, that are generated from templates but written as free text, were evaluated using Skill via Claude. Results were compared using GPT-5.3-codex and an independent review by two pathologists. A local run was done using Mistral 7B via Ollama to show the potential local implementation.

Results

Skill has parsed and identified most of the required parameters, and claimed that 80% of the reports were complete. Pathologists evaluation revealed false positive and false negative assessments. Mesorectum grade assessment for non-rectal specimens, incorrect lymph node count, failure to parse combined pTN notation, unrecognized Turkish terminology were main reasons. Colon resections for non-colorectal tumors were not identified as metastasis and evaluated as missing colorectal report features.

Inter-AI concordance between Claude runs was 49% and 67% (as assessed by each pathologist, respectively).

Local Mistral 7B run 145 seconds/report.

Conclusion

Pathology report quality skill seems to be promising to be used in cloud and local models. Skills can be used as pre-signout check to increase report quality, and can be integrated in routine reporting workflow.

Keywords: large language models, skills, guidelines, pathology report, quality assurance,

Notes:

Author's Index preview

B

Balci Pelin 1
Balci Serdar 1

N

Nazlim Sinan 1

T

Turkmen Ilknur 1

U

Unlu Mehtat 1